KPDS ve YDS – 22 HAFTALIK PROGRAM – PAKET 8

19 Ocak 2004– 25 Ocak 2004

İçerik:

SÖZCÜK BİLGİSİ SORULARI

Bu dosyada yer alan sorular, ***hakangur*** sitesinde yer alan sözcük bilgisi notlarına koşut olarak oluşturulmuştur. Sorular, sözcük bilgisi notlarının altışar konusunu (1-6, 7-12, 13-18, 19-23) içermektedir.

Birinci Test (Konu 1-6)

1. It is true that I'm older than him, but still we are good ......

A) acquaintance B) company C) members D) adversaries E) opponents

2. He felt so sorry that he was rather ..... when he came to see his girlfriend off at the train station.

A) tearful B) unaffected C) indifferent D) active E) encouraging

3. When he was taken to hospital, his whole body was shaking because of ..... and convulsions.

A) seizures B) illnesses C) fitness D) sneering E) repulses

4. We noticed that he had changed the song, but only ......

A) vastly B) greatly C) slightly D) occasionally E) slimly

5. He talked to me so ..... that you would think we were good friends.

A) intimately B) distantly C) independently D) fiercely E) remotely

6. Some alligators need to swallow large stones to ..... their food better.

A) process B) digest C) endeavour D) feed E) contaminate

7. Although grizzly bears ..... in winter, polar bears don't.

A) emigrate B) feed C) hibernate D) give birth E) fatten

8. We have a small ..... in our back garden where we keep our tropical plants in winter.

A) deck B) dwelling C) yard D) farm E) greenhouse

9. The book was so old that its paper started to ..... when I opened its pages.

A) hustle B) rust C) decay D) decompose E) dismiss

10. When we landed safely, I gave out a sigh of ....

A) fear B) worry C) relief D) panic E) comfort

11. Because it is a military area, taking pictures is ....

A) prohibited B) compulsory C) obligatory D) forgiven E) legalized

12. What we need is ..... criticism so that we can form an opinion.

A) disruptive B) constructive C) attentive D) preliminary E) suicidal

13. In order to escape punishment, he ..... insane at the court.

A) demanded B) regretted C) urged D) retained E) pleaded

14. I'm so tired that I can't help ......

A) coughing B) sneezing C) yawning D) giggling E) hiccuping

15. I'm allergic to perfume; it makes me ..... repeatedly.

A) cough B) sneeze C) yawn D) giggle E) hiccup

16. An unexpected bone disease left him ____; now he is confined to bed.

A) injured B) wounded C) unable D) invalid E) pessimist

17. His ..... took shorter than the doctors had expected.

A) recovery B) embedding C) heal D) ailing E) fever

18. He is a devoted member of the club and provides ..... aid for the alcoholic people.

A) reluctant B) disturbing C) pioneering D) voluntary E) optimistic

19. In an accident, speedy transport of the people involved to hospital is a very important factor for their ......

A) survival B) casualty C) prevention D) existence E) extinction

20. Her father is the fattest person you will ever see but Jane is very ....., unlike her father.

A) overweight B) tender C) slender D) obese E) resolute

21. They have built special routes for the physically ..... to travel on.

A) handicapped B) powerful C) tender D) energetic E) healing

22. As she comes from a noble family, she can trace her ..... back to the beginning of the fifteenth century.

A) offspring B) children C) heirs D) descendants E) ancestors

23. A peace ..... between the two fighting sides will help survive millions of refugees.

A) writing B) treaty C) warfare D) treatment E) struggle

24. He ..... attacked the tramp in order to rescue his money and passport.

A) greedily B) disobediently C) rebelliously D) safely E) sincerely

25.In order to understand the motives of the murderer, you need to ..... yourself with the mental situation he is in.

A) interrupt B) acquaint C) realize D) understand E) introduce

İKİNCİ TEST (7-12)

1. They finally ..... to the enemy and were taken as prisoners of war.

A) acknowledged B) approached C) bargained D) spoke E) surrendered

2. When he found several alibis, he managed to prove his own ..... .

A) guilt B) innocence C) naiveté D) optimism E) suspicion

3. When America was a young and undiscovered country, many ..... set out and drew the routes for later comers.

A) discoverers B) fanatics C) inventors D) pioneers E) rioters

4. Thomas went on to complete his project ..... of all the objections he faced.

A) hopeless B) regardless C) tactless D) thoughtless E) unless

5. Who do you ..... will the new president?

A) approximate B) demote C) determine D) promote E) reckon

6. She is such a ..... child that her parents do not know what to do.

A) incoherent B) mischievous C) mishap D) nonsensical E) unintelligible

7. The murderer's ..... for his victim was so strong that he stabbed her several times.

A) adornment B) approval C) detest D) refusal E) rejection

8. I am ..... to know the test result, because I'll apply for a job if I've passed it.

A) careless B) casual C) cautious D) desperate E) hopeless

9. She is so ..... that she will use a toothpick several times.

A) affluent B) cheap C) flamboyant D) generous E) mean

10. The notion that your house-plants will ..... faster if you give them plenty of water is wrong.

A) become B) bring up C) cultivate D) grow E) harvest

11. I don't understand how I manage to ..... such dreadful photos!

A) bring B) carry C) fetch D) hold E) take

12. Some people argue that ..... can well be given at home, without the presence of a teacher.

A) education B) insurance C) learning D) studying E) testing

13. As my husband is very forgetful, I need to ..... him everything.

A) recall B) remember C) remind D) retain E) return

14. He neatly ..... the towel and placed it in the drawer.

A) bent B) folded C) twisted D) wound E) wrapped

15. With such ..... directions, it is a miracle that you managed to find the address.

A) accurate B) amended C) confusing D) thorough E) upset

16. The police have found no apparent ..... of the motives of the assassin.

A) consequence B) exclamation C) indication D) investigation E) sequence

17. His master's thesis ..... that not all children learn the same way.

A) approves B) authorizes C) exploits D) favours E) reveals

18. Any shapeless metal object will ..... when it is put on water surface.

A) drop B) drown C) fall D) sink E) spill

19. His ..... is so deeply rooted that he won't even care about his own children.

A) optimism B) selfishness C) sociability D) tactfulness E) tolerance

20. He is a sturdy person, but even he failed to ..... the severe cold of the place.

A) endure B) enjoy C) feel D) last E) use

21.It is not ..... of you to postpone your studies until the last moment.

A) charitable B) hospitable C) profitable D) reasonable E) respectable

22.The fugitive was last ..... near a barn, but he hasn't been seen since then.

A) appeared B) demonstrated C) flown D) realized E) sighted

23.The Minister of Foreign Affairs has been asked by the Prime Minister to resign because he is very ..... in his relations with foreign ministers of other countries.

A) dedicated B) incompetent C) independent D) inestimable E) meticulous

24.We decided keep the bad news as this would have ..... effects on the pregnant woman.

A) amazing B) boring C) disturbing D) soothing E) tedious

25. Simply out of ....., we decided to play a round of cards although none of us liked card games.

A) anxiety B) boredom C) concern D) enthusiasm E) interest

ÜÇÜNCÜ TEST (13-18)

1. My ..... impression of him was not very favourable; yet, I started to like him as I got to know him.

A) initial B) introductory C) lasting D) permanent E) preparative

2. People are demonstrating against the plan for the ..... of the old church because it is an architectural masterpiece of the 17th century.

A) alleviation B) construction C) demolition D) erecting E) fabrication

3. She has moderate feeling about everything; she is never ..... about anything.

A) passionate B) placid C) relaxed D) serene E) unruffled

4. Even though he is a very serious person himself, he has a strong sense of ..... .

A) common sense B) humour C) logic D) reason E) satire

5. In some countries, women as well as men are ..... as soldiers.

A) attained B) enrolled C) fetched D) recruited E) secured

6. While he gave an account of the brave action he'd taken in the face of danger of death, he talked very ....., as if it was a most usual thing.

A) assuredly B) confidently C) modestly D) positively E) quietly

7. How many times do I have to tell you not to ..... with my plans!

A) dangle B) interfere C) intermit D) interrupt E) suspend

8. Even though he was a prominent writer in the seventies, Kurt Vonnegut failed to ...... much after his masterpiece *Slaughterhouse Five*.

A) accomplish B) bestow C) extend D) reach E) yield

9. He ..... so much on my dress that I started to wonder whether he was serious or not.

A) advocated B) complimented C) elevated D) encouraged E) promoted

10. There were so many ..... during the film due to several commercials that I couldn't keep track of the flow of events.

A) inabilities B) infringements C) interruptions D) intrusions E) inundations

11.The ….. he had brought to the problem-solving mechanism of the company was adorable.

A) banality B) insignificance C) pettiness D) simplicity E) triviality

12.As the meaning of this sentence conveys a surprise rather than a question, you should place not a question mark but an ….. mark to its end.

A) anxiety B) eagerness C) excitement D) exclamation E) outcry

13.I realized that my students had not understood a few grammar points and decided to make a full ….. of these points.

A) issue B) outset C) revision D) start E) topic

14.It may be that Nigeria has a great ….. thanks to its oil wells, but it also true that it is not shared fairly among its people.

A) abundance B) destiny C) generosity D) nobleness E) wealth

15.They all tried to ….. him to come with them, but he would not change his mind.

A) controvert B) deny C) oppose D) persuade E) satisfy

16.Obesity is such that the more overweight you are, the more you want to eat and, ….., the more you eat, the more overweight you get.

A) conversely B) initially C) originally D) thoroughly E) wholly

17.The delegation finally decided that the games should be open ….. to French citizens.

A) directly B) instantly C) simply D) solely E) uniquely

18.Rather than buying ….. coffee, we buy seeds as then it does not lose its original flavour.

A) crushed B) ground C) powdered D) pre-mixed E) pulped

19.In came a young woman who looked ….. , her clothes being to the fashion.

A) affluent B) ragged C) shabby D) valuable E) worn

20.The Prime Minister ….. with himself; on the one hand, he said his government would support free speech and on the other hand he got angry with journalists for asking too many questions.

A) complied B) conformed C) contradicted D) debated E) favoured

21.The service in this hotel ….. everything, even transport to the beach.

A) dissuades B) implies C) includes D) indicates E) proclaims

22.My duties in the office are rather heavy already and I would not like to ..... any more responsibilities.

A) undergeneralize B) undergo C) understate D) undertake E) undervalue

23.You'd better listen to him with utmost care; he generally gives clues with highly ..... value.

A) ambiguous
B) incomprehensible
C) meaningless
D) obscure
E) significant

24.Judging from the information from a previous event, we ..... that over a hundred thousand people will be watching this rock concert.

A) estimate B) execute C) observe D) perform E) regard

25.I am not good at giving ..... because I easily forget what a person looks like.

A) appearances B) descriptions C) expressions D) indications E) visions

DÖRDÜNCÜ TEST (1-23)

1. The bus conductor told him to get off because he couldn't pay the ..... .

A) bill B) fare C) fee D) journey E) travel

2. I felt a sharp ..... when I put my hand in the boiling water.

A) ache B) harm C) hurt D) pain E) suffer

3. I am very fond of Graham Greene's novels. He is my ..... modern author.

A) favoured B) favourite C) likely D) popular E) preferred

4. She chose some attractive ..... paper for the Christmas present.

A) covering B) envelope C) involving D) packing E) wrapping

5. It's rude to interrupt when someone else is ..... .

A) discussing B) remarking C) saying D) talking E) telling

6. Look, Mother! Jack has ..... you some lovely flowers.

A) brought B) carried C) lifted D) present E) taken

7. He ..... out of the window for a moment and then went on working.

A) glanced B) glimpsed C) regarded D) saw E) viewed

8. The company made a record ..... last year.

A) benefit B) earn C) profit D) wage E) winning

9. ..... an experiment of two months, children were given sweets of varying colours.

A) By B) Until C) During D) As E) When

10. These cars originally had two doors but the latest ..... has four.

A) brand B) mark C) model D) pattern E) trade

11. He was killed in a car ..... .

A) blow B) crash C) flash D) hit E) shock

12. He's a nice man but he's ..... to drink too much at parties.

A) adequate B) apt C) common D) probable E) suitable

13. He has a bad cold and won't be ..... to play in the match tomorrow.

A) adequate B) appropriate C) fit D) proper E) suitable

14. He ..... his wife and children and left them to take care of themselves.

A) abandoned B) let C) missed D) spoilt E) wasted

15. We want to make our products cheaper than our ..... .

A) colleagues' B) competitors' C) enemies' D) experts' E) partners'

16. It's the ..... in this country for the father of the bride to pay for the wedding.

A) common B) custom C) habit D) normal E) use

17. He is a very ..... player. He practises for two hours every morning.

A) amateur B) anxious C) excited D) exciting E) keen

18. The bank will ..... you the money if you are prepared to pay them eight per cent interest on it.

A) borrow B) lend C) make D) possess E) put

19. I ..... to him for my bad behaviour.

A) apologised B) coped C) excused D) forgave E) pardoned

20. The sky is ..... . I don't think it will rain.

A) clean B) clear C) cloudy D) open E) tidy

21. I want to see all of you here tomorrow morning at nine o'clock without ..... .

A) fail B) fault C) late D) miss E) neglect

22. He ..... the letter carefully and put it in the envelope.

A) bent B) curved C) folded D) turned E) twisted

23. The price of the meal ..... a service charge.

A) encloses B) enters C) envelopes D) includes E) inspects

24. He shouldn't be allowed to play tennis in the club. He's not a ..... .

A) belong
B) member
C) partner
D) representative
E) social

25. He has always wanted to see his name in ..... .

A) news B) paper C) press D) print E) publication

SÖZCÜK BİLGİSİ

KONU 15

1. ELECTRICAL, FLUID, LIQUID, MASS, MATERIAL, MATTER, PARTICLE

Matter, in science, is a general term applied to anything that occupies space and has the attributes of **gravity** and **inertia**. In classical physics, matter and energy were considered two separate concepts that lay at the root of all physical phenomena. Modern physicists, however, have shown that it is possible to transform matter into energy and energy into matter and have thus broken down the classical distinction between the two concepts. When dealing with a large number of phenomena, however, such as motion, the behaviour of liquids and gases, and heat, scientists find it simpler and more **convenient** to continue treating matter and energy as separate **entities**.

Certain elementary particles combine to form atoms; in turn, atoms combine to form molecules. The properties of individual molecules and their distribution and arrangement give to matter in all its forms various qualities such as mass, hardness, viscosity, fluidity, colour, taste, electrical resistivity, and heat conductivity, among others.

In philosophy, matter has been generally regarded as the raw material of the physical world, although certain philosophers of the school of idealism, such as the Irish philosopher George Berkeley, **denied** that matter exists independently of the mind. Most modern philosophers accept the scientific definition of matter.

1. gravity	a) celestial attraction
2. inertia	b) inactivity
3. convenient	c) reject
4. entity	d) suitable
5. deny	e) thing

2. COLLECT, CONTAIN, LIQUID, MATTER, PARTICLE, POINT, SOLID, SUBSTANCE

In classical physics, there are three forms in which matter occurs—solid, liquid, and gas. Plasma, a collection of **charged** gaseous particles containing nearly equal numbers of negative and positive ions, is sometimes called the fourth state of matter. Solid matter is characterized by resistance to any change in shape, caused by a strong attraction between the molecules of which it is composed. In liquid form, matter does not resist forces that act to change its shape, because the molecules are free to move with respect to each other. Liquids, however, have sufficient molecular attraction to resist forces tending to change their **volume**. Gaseous matter, in which molecules are widely **dispersed** and move freely, offers no resistance to change of shape and little resistance to change of volume. As a result, a gas that is not **confined** tends to **diffuse** infinitely, increasing in volume and diminishing in density.

Most substances are solid at low temperatures, liquid at medium temperatures, and gaseous at high temperatures, but the **states** are not always distinct. The temperature at which any given substance changes from solid to liquid is its melting point, and the temperature at which it changes from liquid to gas is its boiling point. The range of melting and boiling points varies widely. Helium remains a gas down to -269° C (-452° F), and tungsten remains a solid up to about 3370° C (about 6100° F).

1. charged	a) condition
2. volume	b) loaded
3. disperse	c) restrict
4. confine	d) scatter
5. diffuse	e) size
6. state	f) spread, expand

3. MARBLE, SAND, SOIL, STONE, WOOD

Stone is inorganic mineral or soil **concretion** of the earth, of **sedimentary**, igneous, or metamorphic origin, commonly used in building, civil engineering, manufacturing, and art. Some of the building stones are basalt, flint, granite, limestone, marble, porphyry, sandstone, slate, and flagstone. **Ornamental** stones, other than precious stones or gemstones, include alabaster, fluorite, jade, jasper, lapis lazuli, labradorite, and malachite. Mexican onyx marble (stalagmitic aragonite) and Algerian onyx marble, less handsomely coloured, are relatively recent additions to the ornamental stones, as is the jasperized wood of Arizona. In recent years about 83 per cent of the stone used for monuments has been granite, about 17 per cent, and marble.

The **quarrying** of stone in some countries accounts for a large proportion of the land affected by surface mining. In the United States, for example, this is exceeded only by those operations for coal, sand, and gravel **extraction**.

1. concretion	a) contraction
2. sedimentary	b) decorative
3. ornamental	c) deposit
4. quarrying	d) excavation
5. extraction	e) pulling out

KONU 16

1. ART, ARTIFICIAL, ARTISAN, CRAFT, DEVELOP, EMBROIDER, FUNCTION, HANDICRAFT, INDUSTRIALIZATION, MAKE, MANULA, MEANS, METHOD, PRODUCE, TECHNIQUE, TERM, USE, WAY

Crafts (also handicrafts) is the practice of making decorative or functional objects, wholly or partly by hand, and requiring both manual and artistic skill. The term *crafts* also refers to objects made in this way.

Crafts today **predominantly** **comprise** weaving, basketmaking, embroidery, quilting, pottery, woodworking, and jewellery making. They are made both by amateur craftsmen at home, as a hobby with a minimum of equipment, and by professionals with a regular **outlet** for their products.

Crafts are also used in occupational therapy. For example, patients may be taught crafts to strengthen weakened muscles or to help in gaining the use of an artificial limb. Emotionally disturbed people are also taught crafts as an outlet for feelings. Crafts also provide the disabled with an occupation that **diverts** attention from their handicaps. Prisoners-of-war have been known to produce crafts of high quality; a notable example of this is the straw-work marquetry executed by Napoleonic prisoners-of-war in England during the early years of the 19th century.

Crafts are as old as human history. Originally fulfilling **utilitarian** purposes, they are now a means of producing aesthetically **appealing** handmade objects in a world dominated by mechanization and standardization. Among the earliest basic crafts are basketry, weaving, straw-work, and pottery. Nearly every craft now practised can be traced back many hundreds or even thousands of years.

Craftwork formed the basis of town and city economies throughout Europe until the Industrial Revolution in the 19th century. Once items could be mass-produced, however, individual artisans were no longer needed. In reaction to the effects of industrialization, the Arts and Crafts Movement began in England in the late 19th century, led by the designer and social reformer William Morris. The strong interest in crafts throughout the Western world today grew in large part from this movement.

In many parts of the world crafts are still produced as they have been for centuries; Chinese basketry and Indonesian batik are examples. In the southern Appalachian highlands of the United States, basketry and woven goods are made today by much the same methods used by the original **settlers** of the region.

Ethnographic collections in museums throughout the world include examples of **indigenous** crafts and artisanry to document the development of various cultures; art museums with archaeological collections frequently supplement their displays of formal art objects by showing examples of related folk crafts. In addition, special museums of folk art and of crafts have been established to preserve and **display** examples of traditional crafts.

Contemporary craft workers can learn much from studying earlier techniques and designs, as well as the work of their peers. Many other sources are **available** to those interested in learning crafts. Books and magazines on history, techniques, and **innovations** can be found in great number for every craft. Courses are offered by schools and colleges, art schools, craft groups, and other organizations. Membership of a craft association is another source of instruction and **inspiration**. Such associations often sponsor lectures and demonstrations, and they offer the opportunity to share ideas with other members through publications, meetings, and crafts fairs.

1. predominant	a) accessible
2. comprise	b) consist of
3. outlet	c) current
4. divert	d) distract
5. utilitarian	e) exhibit
6. appealing	f) functional
7. settler	g) motivation
8. indigenous	h) native
9. display	i) novelty
10. contemporary	j) pioneer
11. available	k) reigning
12. innovation	l) release
13. inspiration	m) tempting

2. ASSEMBLY LINE, CHANGE, DEMANDING, DEVELOP, ENGAGED, FUNCTION, INDUSTRIALIZATION, MAKE, MANUFACTURE, MIX, NATURAL, OUTPUT, PLANT, PROCESS, PRODUCE, RATE, SHIFT, TECHNOLOGY, TERM, USAGE, USE, UTILIZE

Industry is the long-term changes in types and distribution of global economic activity. In everyday usage, the term “industry” refers to large manufacturing companies, such as the big multinational car companies. Here, a broader definition of industry will be used, which includes all economic activities, in all sectors, and groups them into three broad categories; primary, secondary, and tertiary.

Primary Industries are the industries responsible for the **extraction** of natural resources. They comprise agriculture, hunting, fisheries, forestry, mining, and quarrying. A **distinction** is often drawn between those primary industries concerned with renewable resources such as forests and those concerned with non-renewable ones, such as minerals.

Secondary industries engage in the manufacturing and production of goods. The word “secondary” implies that such companies are engaged in the second stage of economic activity. They use the natural resources of the primary industries (and possibly the **goods** of other secondary industries) to make products. Secondary industries include house-building and the manufacture of clothes, food-processing, shoes, luggage, furniture, packaging, chemicals, metal products, machinery, electrical products, electronic products, computers, cars, trains, and aeroplanes. They also include utilities, which provide services such as gas, water, and electricity.

Tertiary industries comprise those companies involved in services, as opposed to those providing an extractive or manufacturing function. The tertiary category includes **retailers** (of clothes, food, and so on), banks, insurance companies, hotels, restaurants, estate agents, lawyers, doctors, accountants, teachers, golf professionals, and television presenters. Although the word “tertiary” might be thought to imply that these industries have developed recently (for example, computer programming), this is not necessarily the case. Many oil companies own oil rigs, petrochemical plants, and garage retail outlets. In such cases it is difficult to say whether this industrial activity is primary, secondary, or tertiary.

The contrast between primary, secondary, and tertiary industries can also be seen at the country level. Advanced economies such as the United Kingdom have a mix of industries, primary, secondary, and tertiary, but with an **overwhelming** emphasis on the tertiary sector. In contrast, many poorer nations still depend for their livelihoods on primary industries such as minerals or agriculture.

The first stages of the Industrial Revolution occurred in the middle of the 18th century with the early development of the steam engine and textiles manufacture. The next **phase** of the Industrial Revolution gathered pace in the last two decades of the 19th century, with the **expansion** of electric power, the chemical industry, the motor industry, and assembly-line production. It was during the final decades of the 19th century and the early decades of the 20th that the process of industrialization intensified, in terms of the number of industries and the number of countries involved.

Since World War II the existing industries in the developed world have become much more sophisticated in their products and their manufacturing processes. Miniaturization is only one **aspect** of this sophistication. At the same time new technologies have encouraged the creation of many new industries, including jet aircraft, computers and electronics in general, satellites, nuclear power, new composite materials, carbon fibre, robotics, telecommunications, and data-processing equipment. In the latter half of the 20th century there have been continued developments in manufacturing, but also a shift in employment in the advanced economies away from secondary and towards tertiary activities. Over the past 20 years the most advanced economies have been forced to re-evaluate their position in the face of **competition** from newly industrializing economies. Many of the so-called Asian Tiger economies such as Hong Kong, Malaysia, and Singapore, recorded rates of economic growth that are two or three times those of developed countries over the decade 1985-1995.

For businesses, technological improvements in transport and communications have opened up a new world of opportunities. An **illustration** of the **surge** in activity can be seen in the following numerical example. If world trade, output, and foreign direct investment are indexed at 100 in 1970, then by 1993 the output index had reached 180, the trade index 300, and the foreign investment index 550. As a result of these changes, and helping to stimulate them, world currency market **turnover** is now over $1,000 billion per day.

However, there are a number of factors that could, at the very least, slow or possibly even **reverse** the globalization process. Where possible, many companies will attempt to offer their goods or services to a global market, but in many instances these will need to be **tailored** to local demands and tastes. There is also the possibility that certain goods and services will have no appeal in other nations. This could be a result of, for example, the contrasting cultures of Western and Islamic nations.

1. extraction	a) appearance
2. distinction	b) breathtaking
3. goods	c) change
4. retailer	d) customize
5. overwhelming	e) difference
6. phase	f) escalation
7. expansion	g) explanation
8. aspect	h) increasing in size
9. competition	i) merchant
10. illustration	j) profit
11. surge	k) property
12. turnover	l) pulling out
13. reverse	m) rivalry
14. tailor	n) stage

3. ACADEMIC, ART, ARTIFICIAL, ARTISAN, BACKGROUND, CLAY, COMBINE, CORRECT, DESTROY, DEVELOP, EFFORT, EXPERIMENT, FORMULATE, FUNCTION, INDUSTRIALIZATION, INVENT, MANUFACTURE, MEANS, MECHANIC, METHOD, NATURAL, PHYSIOLOGY, PROCESS, PRODUCE, SCHOLAR, SCIENCE, SPECIALIZE, TECHNOLOGY, TERM, USE, WAY

Science (Latin, *scientia,* from *scire,* "to know") is the term used in its broadest sense to **denote** systematized knowledge in any field, but usually applied to the organization of objectively verifiable sense experience. The **pursuit** of knowledge in this context is known as *pure science,* to distinguish it from *applied science,* which is the search for practical uses of scientific knowledge, and from technology, through which applications are realized.

Efforts to systematize knowledge can be traced back to prehistoric times, through the designs that Palaeolithic people painted on the walls of caves, through numerical records that were carved in bone or stone, and through **artefacts** surviving from Neolithic civilizations. The oldest written records of protoscientific investigations come from Mesopotamian cultures; lists of astronomical observations, chemical substances, and disease symptoms, as well as a variety of mathematical tables, were **inscribed** in cuneiform characters on clay tablets. Other tablets dating from about 2000 BC show that the Babylonians had knowledge of Pythagoras' Theorem, solved quadratic **equations**, and developed a sexagesimal system of measurement (based on the number 60) from which modern time and angle units **stem**.

Scientific knowledge in Egypt and Mesopotamia was chiefly of a practical nature, with little rational organization. Among the first Greek scholars to seek the **fundamental** causes of natural phenomena was the philosopher Thales, in the 6th century BC, who introduced the concept that the earth was a flat disc **floating** on the universal element, water. The mathematician and philosopher Pythagoras, who followed him, established a movement in which mathematics became a discipline fundamental to all scientific investigation. The Pythagorean scholars **postulated** a spherical earth moving in a circular orbit about a central fire. In Athens, in the 4th century BC, Ionian natural philosophy and Pythagorean mathematical science combined to produce the syntheses of the logical philosophies of Plato and Aristotle. At the Academy of Plato, **deductive reasoning** and mathematical representation were emphasized; at the Lyceum of Aristotle, **inductive** reasoning and qualitative description were stressed. The **interplay** between these two approaches to science has led to most subsequent advances.

During the so-called Hellenistic Age following the death of Alexander the Great, the mathematician and inventor Archimedes laid the foundations of mechanics and hydrostatics (part of fluid mechanics); the philosopher and scientist Theophrastus became the founder of botany; the astronomer Hipparchus developed trigonometry; and the anatomists and physicians Herophilus and Erasistratus based anatomy and physiology on **dissection**.

Following the destruction of Carthage and Corinth by the Romans in 146 BC, scientific **inquiry** lost its **impetus** until a brief **revival** took place in the 2nd century AD under the Roman emperor and philosopher Marcus Aurelius. At this time the geocentric (earth-centred) Ptolemaic System, advanced by the astronomer Ptolemy, and the medical works of the physician and philosopher Galen became standard scientific treatises for the **ensuing** age. A century later the new experimental science of alchemy arose, springing from the practice of metallurgy. By 300, however, alchemy had acquired an **overlay** of secrecy and symbolism that **obscured** the advantages such experimentation might have brought to science.

During the Middle Ages, six leading culture groups were in existence: the Latin West, the Greek East, the Chinese, the East Indian, the Arabic, and the Mayan. The Latin group **contributed** little to science before the 13th century, the Greek never rose above paraphrases of ancient learning, and the Mayan had no influence on the growth of science. In China, science enjoyed periods of progress, but no **sustained** **drive** existed. Chinese mathematics reached its **zenith** in the 13th century with the development of ways of solving algebraic equations by means of matrices, and with the use of the arithmetic triangle. More important, however, was the impact on Europe of several practical Chinese innovations. These included the processes for manufacturing paper and gunpowder, the use of printing, and the mariner's compass. In India, the chief contributions to science were the formulation of the so-called Hindu-Arabic numerals, which are in use today, and in the conversion of trigonometry to a quasi-modern form. These advances were transmitted first to the Arabs, who combined the best elements from Babylonian, Greek, Chinese, and Hindu sources. By the 9th century, Baghd☐d, on the River Tigris, had become a centre for the translation of scientific works, and in the 12th century this learning was transmitted to Europe through Spain, Sicily, and Byzantium.

Recovery of ancient scientific works at European universities led, in the 13th century, to **controversy** over scientific method. The so-called realists **espoused** the Platonic approach, whereas the nominalists preferred the views of Aristotle. At the universities of Oxford and Paris, such discussions led to advances in optics and kinematics that paved the way for Galileo and the German astronomer Johannes Kepler.

The Black Death and the Hundred Years' War disrupted scientific progress for more than a century, but by the 16th century a revival was well under way. In 1543 the Polish astronomer Nicolaus Copernicus published *De Revolutionibus Orbium Coelestium* (On the Revolutions of the Heavenly Bodies), which revolutionized astronomy. Also published in 1543, *De Corpis Humani Fabrica* (On the Structure of the Human Body) by the Belgian anatomist Andreas Vesalius corrected and modernized the anatomical teachings of Galen and led to the discovery of the circulation of the blood. Two years later the *Ars Magna* (Great Art) of the Italian mathematician, physician, and astrologer Gerolamo Cardano initiated the modern period in algebra with the solution of cubic and quartic equations.

Essentially modern scientific methods and results appeared in the 17^{th} century because of Galileo's successful combination of the functions of scholar and artisan. To the ancient methods of induction and deduction, Galileo added systematic **verification** through planned experiments, using newly invented scientific instruments such as the telescope, the microscope, and the thermometer.

The scientific discoveries of Newton and the philosophical system of the French mathematician and philosopher René Descartes provided the background for the materialistic science of the 18^{th} century, in which life processes were explained on a physicochemical basis.

During the 18^{th} century academies of science were established by other leading nations. The number of scientific journals grew so rapidly during the early 20^{th} century that *A World List of Scientific Periodicals Published in the Years 1900-1933* contained some 36,000 entries in 18 languages. A large number of these are **issued** by specialized societies **devoted** to individual sciences, and most of them are fewer than 100 years old. In addition to national and international scientific organizations, numerous major industrial firms have research departments; some of them regularly publish **accounts** of the work done or else file reports with government patent offices, which in turn print abstracts in bulletins that are published periodically.

1) denote	1. anatomization
2) pursuit	2. assume
3) artefact	3. basic, first
4) inscribe	4. continuous
5) equation	5. cover
6) stem	6. defend
7) fundamental	7. description
8) float	8. disagreement
9) postulate	9. evolve from
10) deductive	10. following
11) reasoning	11. formula
12) inductive	12. interaction
13) interplay	13. introductory
14) dissection	14. investigation
15) inquiry	15. logical, reasonable
16) impetus	16. loyal
17) revival	17. monument, relic
18) ensuing	18. motive
19) overlay	19. motive, pressure
20) obscure	20. peak
21) contribute	21. promote
22) sustained	22. publish
23) drive	23. rebirth
24) zenith	24. represent
25) controversy	25. search
26) espouse	26. swim
27) verification	27. thinking
28) issue	28. unclear
29) devoted	29. validation
30) account	30. write on